Bank Term Deposit Amount Prediction

Regression Modeling and Campaign Targeting Report

Problem Statement

The bank wants to prioritize clients for term deposit marketing campaigns. Since campaign capacity is limited, the goal is to estimate expected deposit value per client and rank clients by predicted value rather than only predicting subscription likelihood.

Dataset and Target Simulation

The dataset was provided during my internship for a banking analytics assignment. Because it does not include actual deposit amounts, a realistic deposit_amount target was simulated. Clients with no subscription receive zero deposit amount; subscribed clients receive positive simulated amounts based on balance, age, job, education, loan indicators, prior campaign behavior, and controlled random noise.

Pre-Campaign Modeling Design

The model is designed for lead prioritization before outreach. The predictive feature set focuses on client profile, account, and campaign history variables available before the contact decision is made. The subscription outcome is used for target construction and validation, while model inputs remain focused on information available at campaign planning time.

Modeling Approach

The notebook compares Linear Regression, Ridge, Lasso, Random Forest, Hist Gradient Boosting, direct XGBoost regression, and a two-stage XGBoost expected-value model. The two-stage model first estimates subscription probability, then estimates positive deposit amount conditional on subscription, and multiplies the two outputs into expected deposit value.

Model Performance

Metric	Value
RMSE	798.45
MAE	359.42
R2	0.3578
Subscription classifier ROC-AUC	0.8078
Subscription classifier PR-AUC	0.4723

Campaign Simulation

Strategy	Clients	Actual Total Deposit	Return / Contact
Top 10%	4,521	8.43M	1,864.37

Top 20%	9,042	10.43M	1,153.84
Random 20%	9,042	2.77M	306.31

The top-20% model-based targeting strategy generated about 3.77x higher actual deposit value than random 20% targeting in the simulation.

Business Recommendation

For a fixed campaign budget that allows contacting 20% of clients, prioritize the top 20% ranked by predicted expected deposit. If the campaign objective is pure efficiency per contact, the top 10% segment is the strongest; if the objective is total deposit capture under the specified budget, the top 20% segment is recommended.

Deliverables

The repository includes an executed final notebook with visible charts and outputs, project documentation, the dataset, requirements, and this summary report.